Unit 10: Hypothesis Testing

10.1 – 10.6: Introduction and Elements of a Test:

Logic, Decisions and Errors

10.3 – 10.5: Large-Sample (*Z-tools*) Tests

Applications: $\mu,\ \mu_1 - \mu_2,\ p,\ p_1 - p_2$

10.8: Small-Sample (*t-tools*) Tests

Applications: $\mu,\ \mu_1 - \mu_2$

10.9: Tests for Variances

Applications: $\sigma^2,\ \sigma_1^2/\sigma_2^2$

10.10: Power and Neyman-Pearson Lemma

10.11: Likelihood Ratio Tests

Hypothesis Testing (Test Set-Up)

Goal: To compare/test two competing claims (hypotheses) about unknown population parameters.

- ***Null*** Hypothesis ($H_0 \rightarrow$ "H naught"):

  This is usually a statement that our parameter(s) are equal to particular value(s), indicating no effect, no difference, no change, no consequence...

- ***Alternative*** Hypothesis ($H_a \rightarrow$ "H - A"):

  This is often a statement that our parameter(s) are in a range of values, indicating some effect, some difference, some change, some consequence...

Example 4:

The amount filled (in ounces) by a certain bottling machine is a random variable following approximately a Normal distribution with unknown mean and standard deviation equal to 2 ounces.

a) Suppose a sample of size n=9 resulted in a sample mean of 14.5 ounces.

Test Question:

Is this an indication that the bottling machine is filling bottles with mean fill less than 16 ounces? Answer the question by carrying out a test.

$\Rightarrow$ Target Parameter: $\mu_Y - average$ fill volume

Test Set-up:

$\Rightarrow$ H_0 : (no) $\mu_Y = 16$ (or you could also write $\mu_Y \geq 16$)

H_a : (yes) $\mu_Y < 16$

Hypothesis Testing (Logic)

- Assume the null hypothesis (H_0) is true.
  The value (associated with the "$=$") in the null hypothesis will be called the hypothesized value.

- Collect evidence (data). Based on the data, measure the strength of the evidence against H_0 in favour of H_a.

Hypothesis Testing (Logic)

Courtroom Trial:

- An individual is on trial because they are accused of being guilty.

$\Rightarrow H_0 : innocent$ vs. $H_a : guilty$

- The trial is carried out under the assumption of innocence (H_0 is true).

- The trial measures the evidence against innocence (H_0) in favour of guilt (H_a).

In Statistical Tests it is the value of the parameter that is on trial!

Decisions/Conclusions:

Based on the evidence/data in the sample you can:

1. Reject H_0 in favour of H_a.

   You found enough evidence against H_0 in favour of H_a.

OR

2. Fail to reject H_0 in favour of H_a.

   You did not find enough evidence against H_0 in favour of H_a.

Example 4: The amount filled (in ounces) by a certain bottling machine is a random variable following approximately a Normal distribution with unknown mean and standard deviation equal to 2 ounces.

a) Suppose a sample of size n=9 resulted in a sample mean of 14.5 ounces. Is this an indication that the bottling machine is filling bottles with mean fill less than 16 ounces?

H_0 : (no) $\mu_Y = 16$

H_a : (yes) $\mu_Y < 16$

$\Rightarrow$ Should we reject or not reject H_0 ?

Is there enough evidence in the sample to conclude that $\mu_Y < 16$?

Does $\bar{y} = 14.5$ from a sample of $n = 9$ imply $\mu_Y < 16$?

Decision Errors:

1. Type I Error: Rejecting a true H_0.

   e.g. Convicting an innocent person.

   Define:
   - $\alpha = P(\text{Type I Error}) = P(\text{Reject } H_0 \mid H_0 \text{ is true})$.
   - α is called the significance level of the test.

2. Type II Error: Failing to reject a false H_0.

   e.g. Failing to convict a guilty person.

   Define: $\beta = P(\text{Type II Error}) = P(\text{Fail to reject } H_0 \mid H_0 \text{ is false})$.

Note: How we ultimately make our decision is based on what we (arbitrarily) deem to be acceptable error probabilities.

Example 4: The amount filled (in ounces) by a certain bottling machine is a random variable following approximately a Normal distribution with unknown mean and standard deviation equal to 2 ounces.

a) Suppose a sample of size n=9 resulted in a sample mean of 14.5 ounces. Is this an indication that the bottling machine is filling bottles with mean fill less than 16 ounces?

H_0 : (no) $\mu_Y = 16$

H_a : (yes) $\mu_Y < 16$

Should we reject or not reject H_0 ?

$\Rightarrow$ Consider our error probabilities:

- $\alpha = P(\text{concluding } \mu_Y < 16 \mid \mu_Y = 16)$.
- $\beta = P(\text{failing to conclude } \mu_Y < 16 \mid \mu_Y < 16)$.

Rejection Region (RR) Approach:

For some function of the sample, $U = g(Y_1, \ldots, Y_n)$, choose to reject H_0 in favour of H_a if $U \in RR$.

e.g. $\cdot\, RR = \left\{ U = \bar{Y} : \bar{y} < 15 \right\}$

$\cdot\, RR = \left\{ U = \prod_{i=1}^{n} Y_i : \prod_{i=1}^{n} y_i > k \right\}$

$\cdot\, RR = \left\{ U = Y_{(k)} : 1 < y_{(k)} < 2 \right\}$

Then: $\alpha = P(U \in RR \mid H_0$ is true),

$\beta = P(U \notin RR \mid H_0$ is false).

Example 4:

H_0 : (no) $\mu_Y = 16$

H_a : (yes) $\mu_Y < 16$

Rejection Region Approach: Since $\bar{Y}$ is the MVUE for μ_Y, consider rejecting H_0 in favour of H_a if we observe $\bar{y} \le k$, for some constant k.

$$\Rightarrow RR = \left\{ \bar{Y} : \bar{y} \le k \right\}.$$

Then,

$$\alpha = P(\text{concluding } \mu_Y < 16 \mid \mu_Y = 16) = P(\bar{Y} \le k \mid \mu_Y = 16),$$

$$\beta = P(\text{failing to conclude } \mu_Y < 16 \mid \mu_Y < 16) = P(\bar{Y} > k \mid \mu_Y = \mu_a < 16).$$

Example 4:

H_0 : (no) $\mu_Y = 16$

H_a : (yes) $\mu_Y < 16$

$$RR = \left\{ \bar{Y} : \bar{y} \le k \right\}.$$

$$\Rightarrow \alpha = P(\bar{Y} \le k \mid \mu_Y = 16) =$$

Questions:

- For fixed k, what is α? Try $k = 13,\ 15.5.$

- For fixed α, what is k? What is your decision?

  Try $\alpha = 0.01,\ 0.05,\ 0.10.$

P-value Approach:

$\Rightarrow$ ***p - value*** $= P($that another sample would produce just as much or more evidence against H_0 in favour of H_A than what was actually observed, if H_0 is true$)$.

- The ***p - value*** is used as a measure for the strength of the evidence against H_0 in favour of H_A.

- The smaller this probability, the stronger the evidence against H_0 in favour of H_a.

Example 4:

H_0 : (no) $\mu_Y = 16$

H_a : (yes) $\mu_Y < 16$

$\Rightarrow$ ***p - value*** $= P($that another sample would produce just as much or more evidence against H_0 in favour of H_A than what was actually observed, if H_0 is true$)$

$=$

Making Decisions (Significance Level Appraoch)

1. Choose a significance level α.
2. Sample and calculate the $p\text{-}value$.
3. Then draw your conclusion:

i) If $p\text{-}value \leq \alpha$ then we say that we have significant evidence against H_0 in favour of H_a at significance level α.

With a significance level of α, we can reject H_0 in favour of H_a.

OR

ii) If $p\text{-}value > \alpha$ then we say that we have insignificant evidence against H_0 in favour of H_a at significance level α.

With a significance level of α, we fail to reject H_0.

Example 4:

H_0 : (no) $\mu_Y = 16$

H_a : (yes) $\mu_Y < 16$

$\Rightarrow$ ***p - value*** ≈ 0.0122.